

AgiBot X2 Ground Recovery

Simulation, reinforcement learning and ROS 2 integration | 27 September 2026

Result: 5/5 supine recoveries. The selected PPO policy starts from random-weight training with every episode supine. All five 10-second validation episodes end in stable two-foot stance without other-body support. No upright-start pretraining or lift assistance is used in this lineage.

Robot and simulation

Item	Implementation
Model	Official AgiBot X2 Ultra v1.3.0 URDF, upstream revision 60c5de5; locally converted to USD.
Physics	Isaac Sim 6.0.1 / Isaac Lab 3.0.0-beta2.patch1 / PhysX; floating base, 200 x 200 m flat floor, self-collision.
Constraints	31 movable joints and 32 sensed bodies; vendor meshes, inertia, joint axes and actuator limits. Position commands use imported 98% soft limits.
Reset / rate	Supine pelvis height 0.190 m, neutral joints and zero velocity; collision clearance checked. Physics 200 Hz, policy 50 Hz; 10 s or 500 policy steps.

Fetch and import: `scripts/fetch_agibot_model.sh` and `scripts/import_x2_isaac.sh`. The generated USD stays local. Mesh collision shapes use the converter's convex-hull approximation.

Observation and action contract

Policy input	Count
Pelvis height; body-frame linear/angular velocity; projected gravity	1 + 3 + 3 + 3
Joint position / velocity	31 + 31
Left/right foot contact; whole-body ground-contact bits	2 + 32
Previous two policy actions	16
Total	122

Eight symmetric channels command hip pitch, knee, ankle pitch, shoulder pitch, elbow, waist pitch, ankle roll and hip roll. They vary 15 joint targets using $\text{centre} + \text{span} * \tanh(\text{action})$, then clamp to imported soft limits. Separate normalized actor/critic MLPs use 512/256/128 ELU layers.

The other 16 joints retain default targets and remain simulated and published: hip yaw; shoulder roll/yaw; wrist roll/pitch/yaw (both sides); waist roll/yaw; head pitch/yaw. Actions specify positions, not direct torques.

Evaluation seeds

One checkpoint, five reproducible supine starts. Each seed samples a different pose within the same limits; seed IDs are local choices.

Seed	Position variation	Orientation variation	Unchanged
101	X/Y: +/-4 cm Z: +/-1 mm	Roll/pitch: +/-0.005 rad Yaw: +/-0.02 rad	Neutral joints; zero velocity.
102	X/Y: +/-4 cm Z: +/-1 mm	Roll/pitch: +/-0.005 rad Yaw: +/-0.02 rad	Neutral joints; zero velocity.
103	X/Y: +/-4 cm Z: +/-1 mm	Roll/pitch: +/-0.005 rad Yaw: +/-0.02 rad	Neutral joints; zero velocity.
104	X/Y: +/-4 cm Z: +/-1 mm	Roll/pitch: +/-0.005 rad Yaw: +/-0.02 rad	Neutral joints; zero velocity.
105	X/Y: +/-4 cm Z: +/-1 mm	Roll/pitch: +/-0.005 rad Yaw: +/-0.02 rad	Neutral joints; zero velocity.

Reward design and PPO experiment

Signed height guides the rise without rewarding an inverted bridge. Transitional hand/body contacts remain allowed. Dense posture feedback and motion/command regularization guide the transition from rising to sustained foot-only stance.

Term	Weight	Purpose / implementation
Signed pelvis height	40	$\text{clip}(z/0.68) * \text{clip}((1-g_z)/2)$.
Head height / upright	5 / 5	Clipped exponential head height; $\exp(-g_z)$.
Feet / other support	10 / -1	Two-foot contact; discourage other support near standing.
Balance / strict stance	20 / 20	Low velocity near standing; complete physical success predicate.
Stance proximity	20	Exponential proximity to stance. Training widths: tilt 0.35, speed 0.8 m/s and angular speed 2 rad/s.
Leg pose	20	Neutral hip pitch/knee/ankle pitch; gated above 0.40 m and by upright orientation.
Relaxed arms	40	Supported upright shoulder pitch 0 and elbow -0.15 rad.
Near-stance motion	-2	Penalize squared base speed and 0.1 times squared angular speed near standing.
Command saturation	-0.5	Squared excess outside effective raw-action limits; accounts for tanh and imported soft-limit clipping.
Action change / joint speed	-0.02 / -0.0005	Squared successive raw-action changes / joint velocities.
Torque / limit / failure	-1e-6 / -1 / -10	Squared effort, soft-limit violation, non-timeout safety termination.
Shoulder command	80	Supported-upright gate * $\exp(-(a_{sh}-\text{atanh}(0.25))^2/v)$; $v=9$, then 1 for final precision.

RewardManager sums weight x term x 0.02 s. HumanUP [1] motivates rise/control terms; HoST [2] motivates post-task posture; FRASA [3] motivates compact symmetric control and desired-pose proximity. Weights, gates, widths and raw-command penalties are X2 choices, not copied paper constants.

Training configuration

PPO [4]: clip 0.2, gamma 0.99, GAE lambda 0.95; 5 epochs, 4 minibatches, value coefficient 1, gradient clip 1, desired KL 0.01. All stages: 3,000 environments, 32 steps/environment, seed 47, entropy 0. Initial std 0.8; actor, critic, optimizer and learned std are retained on resume. Selected updates: 500 + 500 + 300 + 454 + 300 + 200 = 2,254. First 1,000: adaptive LR starting at 3e-4; then 1e-4 fixed, finally KL-adaptive. The final 200 updates narrow the shoulder reward. Repeated resume indices explain the filename model2248.



Complete selected ancestry; faint: logged return, dark: 20-update mean. Markers show resumes and reward changes after 1,000 and 2,054 updates; return scales differ. The rejected branch is recorded separately.

Five-episode validation

A recovery requires every condition below for at least 0.5 continuous seconds. Evaluation continues for the complete episode; terminal samples are captured before automatic reset. Final stance duration is reported separately from a brief earlier success.

Check	Threshold
Height	Pelvis at least 0.58 m.
Orientation	$\text{norm}(\text{projected_gravity_xy}) \leq 0.15$ and $g_z \leq -0.98$.
Base motion	Linear speed ≤ 0.25 m/s; angular speed ≤ 0.35 rad/s.
Support	Each foot ≥ 15 N ground force; every other body < 15 N.
Additional arm check	Shoulder pitch and elbow target errors ≤ 0.30 rad during strict stance; this is additional to the HRS recovery criterion.

Measured outcome

Seed	Recovery	Strict stance at end	Strict + relaxed arms at end
101	Yes	9.12 s	8.50 s
102	Yes	8.94 s	8.40 s
103	Yes	9.14 s	8.54 s
104	Yes	8.94 s	8.52 s
105	Yes	9.06 s	8.52 s

5/5 successful recoveries. All five end on both feet without other-body support, with 8.94-9.14 s of uninterrupted strict stance at episode end. All five also satisfy the neutral-arm check throughout the final two seconds. Maximum shoulder/elbow errors: 0.233/0.099 rad. Largest arm-joint position range: 0.00195 rad; velocity RMS: 0.0261 rad/s (50 Hz).

Source: reports/relaxed_v4_evaluation.json. Checkpoint: reports/checkpoints/x2_supine_model2248.pt.

Failure analysis and scope

The original supine-only 500-update run scored 0/5. It reached 0.763-0.783 m while airborne and never obtained strict-stance reward. Earlier successful weights had mixed-start pretraining, which supplied early balance experience; that lineage is separate from this result.

The new lineage retains every supine reset while adding dense stance/leg-pose feedback and motion/action regularization. The first 500 updates still scored 0/5; continuing the same preset reached 5/5 after 1,000 updates. This is a combined intervention and larger training budget, not an ablation proving one reward alone caused the improvement.

Shoulder-command shaping then reduced the forward-arm pose. A late 146-update fixed-rate branch regressed and was rejected. Resume from the earlier stable parent with KL-adaptive learning rate preserved balance; narrowing the shoulder reward from denominator 9 to 1 met the unchanged arm tolerance. All starts remained supine.

A separate random-weight run with the complete final reward combination fixed from the start scored 0/5 after 500 updates (48 million steps). It rose to 0.718-0.762 m but never satisfied all strict stance conditions together. This is one training seed, not a universal minimum-budget result.

Scope: nominal flat-floor simulation with small reset perturbations. Terrain variation, broad pose recovery, sensor noise, latency and real-hardware performance were not validated. Development failures and exact configuration provenance remain in Git and docs/development_history.md.

ROS 2 interface and reproduction

The Python package builds with colcon and launches recovery and telemetry nodes together. A local Unix socket connects ROS 2 Humble (Python 3.10) to the Isaac policy server (Python 3.12). Joint positions come from the live simulator.

Interface	ROS 2 type / behavior
/x2/start_recovery	std_srvs/srv/Trigger: acceptance before execution; a second request while busy returns false.
/x2/recovery_status	std_msgs/msg/String: IDLE, RUNNING, SUCCEEDED or FAILED.
/x2/joint_states	sensor_msgs/msg/JointState: timestamps and 31 measured joint names/positions while running.

Telemetry logs status and left_knee_joint at 1 Hz. ROS ends after the physical 0.5 s success hold; the separate full-length evaluation verifies final arm posture. Timeout is configurable. A fresh isolated colcon build passed; the selected exported policy was tested through the real ROS-Isaac connection.

Recorded check	Observed outcome
CLI recovery	RUNNING to SUCCEEDED; 78 timestamped 31-joint samples.
Concurrent request	First accepted in 1.08 ms; second rejected while busy.
Timeout 0.2 s	RUNNING to FAILED; 10 joint samples; simulator timeout reported.

Commands from the repository root

Install the pinned dependencies using README.md and docs/commands.md. Training is optional for supplied-checkpoint playback.

```
export ISAAC_PYTHON=/path/to/isaac/environment/bin/python
./scripts/fetch_agibot_model.sh
./scripts/import_x2_isaac.sh
CKPT=reports/checkpoints/x2_supine_model2248.pt
./scripts/play_isaac.sh "$CKPT" --environment relaxed_v4 \
  --device cuda:0 --viz kit --start-delay 2 --seeds 101 \
  --output /tmp/x2_play.json
./scripts/evaluate_isaac.sh "$CKPT" --environment relaxed_v4 \
  --device cuda:0 --output /tmp/x2_evaluation.json
```

The two-second preview freezes the initial pose; physics and the episode clock do not advance. Seeds select small pose perturbations. Each new training run saves its checkpoint, reward PNG/CSV and manifest under its own timestamped logs/rsl_rl directory.

For ROS: build with scripts/build_ros.sh, start the Isaac policy server, launch both ROS nodes in a separately prepared terminal, then call /x2/start_recovery. Ordered commands and expected outputs are in docs/commands.md and docs/ros_live_validation.md.

Evidence and references

Evidence: reports/ (weights, training graph, evaluation and ROS logs); docs/parameter_provenance.md (parameter sources); docs/development_history.md (failures and fixes). Tested host: Ubuntu 22.04.5, RTX 5080 Laptop GPU with 16 GB.

[1] HumanUP, RSS 2025. doi:10.15607/RSS.2025.XXI.063
[2] HoST, RSS 2025. doi:10.15607/RSS.2025.XXI.064
[3] FRASA, arXiv:2410.08655, Section IV-D
[4] Proximal Policy Optimization, arXiv:1707.06347